JOHANSSON ENGINEERING

Risk Assessment — Pressure Vessel Maintenance Programme

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1. Summary

This risk assessment identifies and evaluates hazards associated with the ongoing maintenance and inspection of pressure vessels at the Göteborg Workshop. It covers normal operations, inspection activities, and foreseeable failure modes. Controls are proposed following the hierarchy of elimination, substitution, engineering controls, administrative controls, and PPE.

2. Scope

- All pressure vessels registered in JDS-LOG-MEC-003
- Inspection and maintenance activities per AFS 2017:3
- Contractor and in-house maintenance personnel
- Normal operation, shutdown, start-up, and emergency scenarios

3. Methodology

Risk is assessed using a 5×5 matrix per ISO 31000 and SS-EN 16991:

Likelihood (L):

SCORE	DESCRIPTION	FREQUENCY
1	Rare	Less than once per 20 years
2	Unlikely	Once per 10–20 years
3	Possible	Once per 5–10 years
4	Likely	Once per 1–5 years
5	Almost certain	More than once per year

Consequence (C):

SCORE	DESCRIPTION	IMPACT
1	Negligible	No injury, minor repair
2	Minor	First aid, localised damage
3	Moderate	Medical treatment, partial shutdown
4	Major	Serious injury, extended shutdown
5	Catastrophic	Fatality, facility loss

Risk Rating = L × C

RATING	CATEGORY	ACTION REQUIRED
1–4	Low	Monitor, no immediate action
5–9	Medium	Controls required within 3 months
10–15	High	Controls required within 1 month
16–25	Critical	Stop work, immediate action

4. Risk Register

4.1 Operational Hazards

ID	HAZARD	CAUSE	CONSEQUENCE	L	C	RISK	CONTROLS	RESIDUAL
R-001	Overpressure event	Relief valve failure, blocked outlet	Vessel rupture, projectiles, blast	2	5	10 HIGH	Relief valve testing per schedule, pressure switch backup (PS-001), operator training	4 MED
R-002	Corrosion-induced wall thinning	Internal corrosion, condensate accumulation	Leak or burst at weakened point	3	4	12 HIGH	Internal revision inspections per AFS 2017:3, UT thickness measurements, drain condensate weekly	6 MED
R-003	Steam boiler dry firing	Low water level, level gauge failure	Boiler tube failure, steam release	2	5	10 HIGH	Low water cutoff device, daily water level checks, annual boiler inspection by DNV	4 MED
R-004	Hydraulic accumulator bladder failure	Fatigue, over-cycling	Sudden pressure loss, press malfunction	3	3	9 MED	Pre-charge pressure checks quarterly, bladder replacement at 5-year interval	4 LOW
R-005	Nitrogen cylinder bank leak	Fitting failure, impact damage	Oxygen displacement in confined gas store	2	4	8 MED	O_2 monitoring alarm in gas store, ventilation, restricted access, cylinder restraints	4 LOW

4.2 Maintenance Activity Hazards

ID	HAZARD	CAUSE	CONSEQUENCE	L	C	RISK	CONTROLS	RESIDUAL
R-006	Stored energy release during maintenance	Incomplete depressurisation, isolation failure	Burns, impact injury	2	4	8 MED	Lock-out/tag-out procedure, double block and bleed, pressure gauge verification at zero	3 LOW
R-007	Confined space entry (vessel internal inspection)	Oxygen depletion, toxic atmosphere	Asphyxiation, poisoning	2	5	10 HIGH	Confined space permit, gas testing, standby person, rescue plan, ventilation	4 MED
R-008	Lifting injury during safety valve removal	Heavy components, awkward access	Musculoskeletal injury	3	2	6 MED	Mechanical lifting aids, two-person lift policy, pre-task briefing	2 LOW
R-009	Chemical exposure during cleaning	Descaling agents, degreasing solvents	Skin/eye irritation, fume inhalation	3	2	6 MED	SDS review, PPE (gloves, goggles, respirator), local exhaust ventilation	2 LOW

5. Risk Matrix Summary

	NEGLIGIBLE (1)	MINOR (2)	MODERATE (3)	MAJOR (4)	CATASTROPHIC (5)
Almost certain (5)					
Likely (4)					
Possible (3)		R-008, R-009	R-004	R-002	
Unlikely (2)				R-005, R-006	R-001, R-003, R-007
Rare (1)					

6. Conclusions

- **3 high risks** identified (R-001, R-002, R-003, R-007) — all related to catastrophic pressure release or confined space work
- All high risks have been reduced to medium or low through existing and proposed controls
- **No critical (stop work) risks** identified
- The maintenance programme as designed in JDS-PRJ-MEC-001 adequately addresses the identified risks

7. Recommendations

PRIORITY	ACTION	RESPONSIBLE	TARGET DATE
High	Install O_2 depletion alarm in gas store (R-005)	Facility manager	2026-Q2
High	Develop confined space rescue plan (R-007)	Safety officer	2026-Q2
Medium	Establish UT thickness baseline for PV-001 and PV-004 (R-002)	Inspection contractor	2026-Q3
Medium	Procure mechanical lifting aid for valve workshop (R-008)	Workshop supervisor	2026-Q3
Low	Review and update SDS file for all maintenance chemicals (R-009)	Nils Johansson	2026-Q4

Revision History

REV	DATE	AUTHOR	DESCRIPTION
A	2026-03-25	Nils Johansson	Example risk assessment for JDS demonstration